Requirement Document: N8N Hands-On Beginner Course

Topic 9: Google Sheets Integration

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 9 - Google Sheets Integration
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
Last Updated	June 24, 2026

2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to connect n8n to Google Sheets to read, append, and update rows in a spreadsheet. Trainees will learn how to configure authentication, map workflow fields to spreadsheet columns, specify target spreadsheets and tabs, and use key columns to update existing records.

2.2 Learning Scope

In Scope:

- Connecting the Google Sheets node in n8n with target spreadsheets.
- Configuring the Append operation to insert new rows of data.
- Configuring the Read operation to retrieve all rows from a spreadsheet tab.
- Configuring the Update operation to find an existing row using a lookup key column (e.g., Email) and modify a specific cell value (e.g., Status).

- Mapping dynamic input variables (e.g., `customerName`, `customerEmail`) to spreadsheet columns (`Name`, `Email`, `Status`).
- Specifying Spreadsheet IDs and Sheet Name tabs (e.g., `Sheet1`).

Out of Scope:

- Creating new spreadsheets or deleting existing sheets dynamically via the API.
- Writing complex cell formulas or modifying cell formatting (font style, cell borders, background colors) via n8n.
- Managing Google Drive folder permissions or sharing spreadsheets dynamically.

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 8.
- A Google Account with access to Google Sheets.
- A pre-configured Google Sheets OAuth2 or Service Account credential loaded in their n8n instance.
- A test spreadsheet created with columns: `Name`, `Email`, `Status`.

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Credentials Config	The workflow shall connect to Google Sheets using valid n8n Google Sheets API credentials (either Service Account or OAuth2).	**High**
REQ-002	Target Specification	The Google Sheets nodes shall require specifying a unique Document (Spreadsheet ID) and a Sheet (tab name, e.g. `Sheet1`).	**High**
REQ-003	Append Operation	The workflow shall append a new row to the sheet, mapping workflow properties `customerName` to the `Name` column and `customerEmail` to the `Email` column, setting the `Status` column to `Pending`.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-004	Read Operation	The workflow shall read rows from the target spreadsheet tab and return the content as standard JSON objects to the workflow.	**High**
REQ-005	Update Operation	The workflow shall update an existing row in the sheet using a lookup: • Lookup Column: `Email` • Lookup Value: `{{ $json.customerEmail }}` • Field to Update: `Status` set to `Verified`.	**High**
REQ-006	Data Pipeline Sync	The workflow shall execute sequentially (Trigger -> Mock Data -> Append -> Read -> Update) and return successful indicators for all steps.	**High**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	API Endpoint Routing	The system must route requests securely to the Google Sheets API (`sheets.googleapis.com/v4/spreadsheets`).
TR-002	Column-Header Parsing	The Google Sheets node must retrieve the first row of the sheet to map incoming keys dynamically to spreadsheet column headers.
TR-003	Row Lookup Resolution	The Update operation must support scanning the specified lookup column for the lookup value and calculating the correct sheet row index to write updates back.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn how to store, query, and synchronize business data between n8n workflows and spreadsheets, which are standard databases in beginner applications.
Instructor / Reviewer	Evaluation	Needs to verify that the trainee knows how to configure the Google Sheets node, map fields correctly, and handle Lookups for update actions.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The trainee has created a Google Spreadsheet and shared access with the n8n Service Account email (if using Service Account auth).
- **Assumption 2 (ASM-002):** The header row in the Google Sheet is row 1 and contains exactly the case-sensitive column titles `Name`, `Email`, and `Status`.
- **Constraint 1 (CON-001):** The Google API limits write frequencies, so bulk operations must respect standard rate limiting constraints.